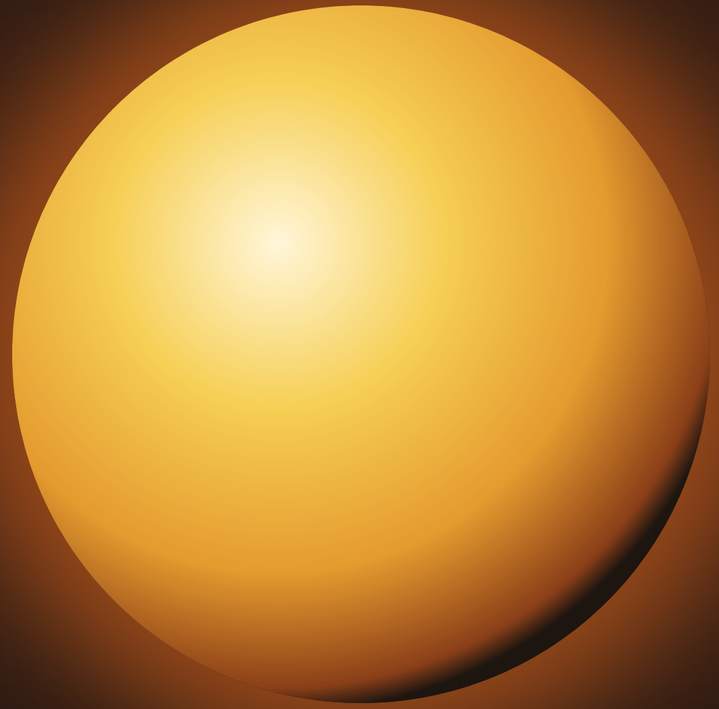
WATT THE HACK · MUNICH · 24 OCT 2026 · NUUENERGY CHALLENGE

Would a heat pump *work* in my home?

Heat Pump Ready? An honest first answer for homeowners, by voice or app, and a smarter visit decision for nuuEnergy's sales team.



Team NoIdea

Track: Buildings & Everyday Energy

01 · THE PROBLEM

Millions of homes still burn gas or oil. Most owners never get past *the first question.*

Homeowner

“Would it even work here?”

Many wrongly assume they need floor heating or a full renovation.

Today

Only a visit answers it

A heating master spends about two hours on site to calculate the heat load.

nuuEnergy

Costly free visits

Some end in “not without major changes”; some homeowners never buy.

02 · WHAT NUUENERGY ASKED

An honest first answer in under three minutes.

From what the homeowner already has: energy bill, building facts, a few phone photos.

- **Green** · likely suitable
- **Yellow** · checks remain
- **Red** · not enough for a yes

“I can’t tell from this data” beats a confident wrong number.

We don’t just tell homeowners if they’re ready. We tell nuuEnergy *which visits are worth paying for.*

For homeowners: one question at a time, by voice or app, in their language. Green, yellow or red with the reason, what’s still open and a rough cost.

For sales: every lead ranked into nuuEnergy’s own next steps, with a briefing that makes the on-site visit shorter.

AI talks. *Rules decide.*

1 · Homeowner

Voice call, app or web

ElevenLabs voice, 8 European languages, photo link by SMS.

→

2 · Orchestrator

Astra + a guard

Asks one question at a time; blocks invented numbers or verdicts.

→

3 · Engine

Deterministic rules

Every number and every verdict, from nuuEnergy's guardrails.

→

4 · Sales desk

Live leads + briefing

With consent: next step, map, and a page for the heating master.

One engine everywhere: web app, phone app, voice agent and sales dashboard call the same tested code. 27 tools, also exposed over MCP.

Three reference houses, *three matching verdicts.*

HOUSE	OUR ESTIMATE	NUUENERGY REFERENCE	OFF BY	VERDICT (EXPECTED)
B-01	9.8 kW	9.6 kW	+2%	Green (green)
B-02	9.9 kW	9.7 kW	+2%	Green (green)
B-03	16.6 kW	14.5 kW	+14%	Yellow (yellow)

Two heat-load methods (floor area × age band, and the real bill) combined by a few named rules, e.g. “a wood stove means the bill is only a lower bound”. No machine learning, no hidden constants: the same input always gives the same answer.

06 · HONEST BY DESIGN

Rules a homeowner can trust, *with sources.*

Missing data is never a “no”

Unknown answers lead to yellow and a technician check, never red.

Building age never rules a home out

Fraunhofer ISE found no correlation between construction year and heat-pump efficiency.

55 °C is a guide, not a wall

Verbraucherzentrale’s rule of thumb; well-sized radiators can work.

Single-pipe means a closer look

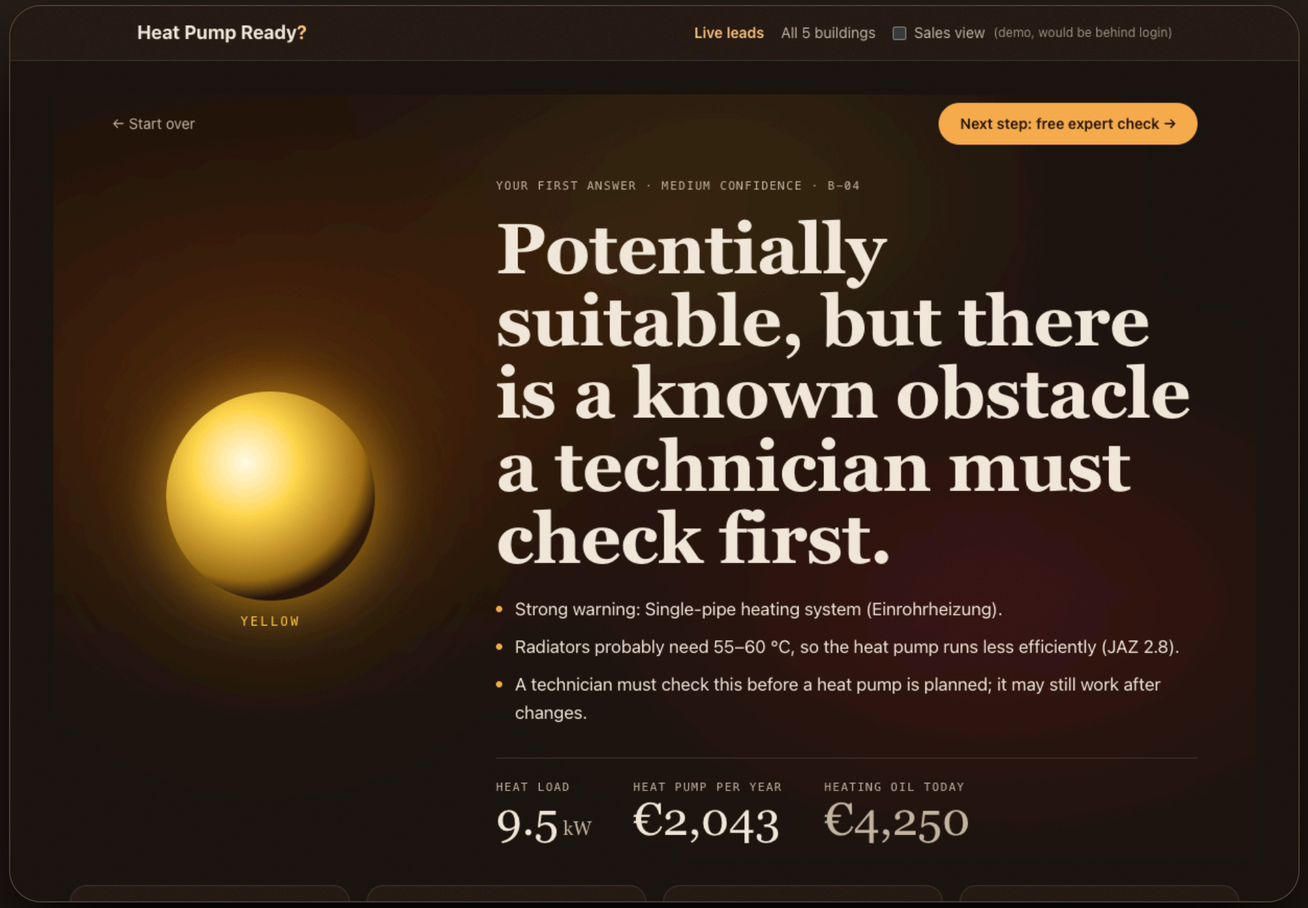
A strong flag with ways forward, as nuuEnergy’s own site describes.

07 · THE ANSWER, IN FOUR PARTS

Every result explains itself.

- **Heat demand** · area, insulation, real use
- **Heating-system fit** · flow temperature, pipes, comfort
- **Installation feasibility** · space, electrics, permissions
- **Information confidence** · what's known, what's open

Each with a status, the reason and the next check: always something to do, never a dead end.



B-04 and B-05: *both yellow, for different reasons.*

B-04 · 1987 · 160 m² · oil

9.5 kW

Yellow: confirmed single-pipe heating, a closer look, not a no.

Heat pump **€2,043** a year vs oil €4,250

B-05 · 2006 · 125 m² · gas

10.4 kW

Yellow: the data contradicts itself: 216 kWh/m² for a 2006 house, 14 rooms in 125 m².

Heat pump **€1,765–2,206** a year vs gas €2,970

Yellow isn’t the end. *It’s a to-do list.*

Today
B-04 · yellow, medium

→

Check
55 °C test passes

→

Check
Radiators sized

→

Bigger work
Loop fixed → green, high

The engine re-runs itself with each technician check assumed positive, cheapest first. B-01 reaches green with checks alone; B-03 and B-05 honestly stay yellow until insulation lowers their 83 W/m².

Just call. *One question at a time.*

Patient by design

Slower calm voice, numbers read back, “I don’t know” always fine, a human on request.

Eight languages

German, English, French, Italian, Spanish, Dutch, Polish, Portuguese, switching mid-call.

Asks only what matters

~9 core questions; more only if the answer would change the result.

Live test calls
2 of 2 correct

Guard had to step in
0 times

Never asked
age · income · bank

AI reads the photos. *It never sets the verdict.*

HELPER	WHAT WE MEASURED	SAFETY RULE
Radiator photo	4 of 5 installer photos right, 0 confidently wrong	Below 70% → “unclear”
Bill reader	Real SWM bill: gas 12,945 kWh, €765.02 correct	Caller confirms first
Meter cabinet, outdoor spot	Real photos: 2 of 4 exact, 0 confidently wrong	A technician confirms
Plain-language summary	5 of 5 kept every number and verdict	New numbers → rejected

Each photo check takes three independent looks and votes. When they disagree, the honest answer is “a technician checks”, which is exactly what the jury asked for.

Every lead lands in *nuuEnergy's own funnel.*

A

Book the on-site visit

~2-hour heat-load visit by the heating master.

B

30-min call first

Their existing technical planning call.

C

Ask for info first

Contradictions, missing photos, approvals.

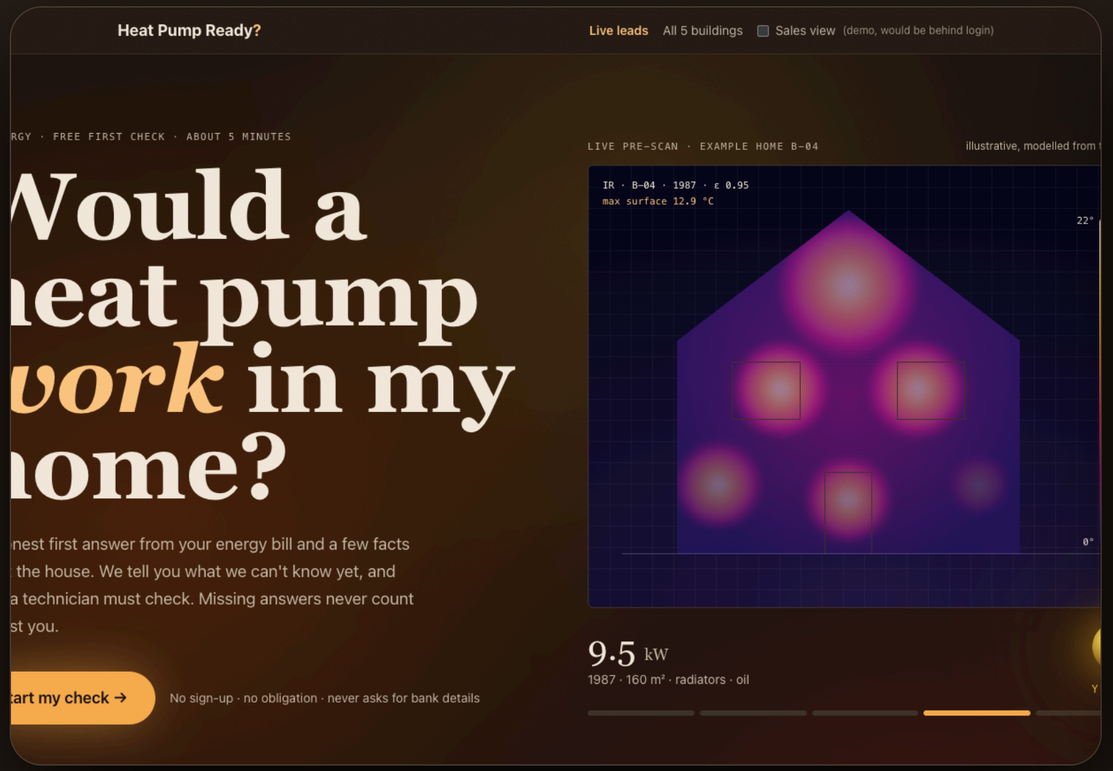
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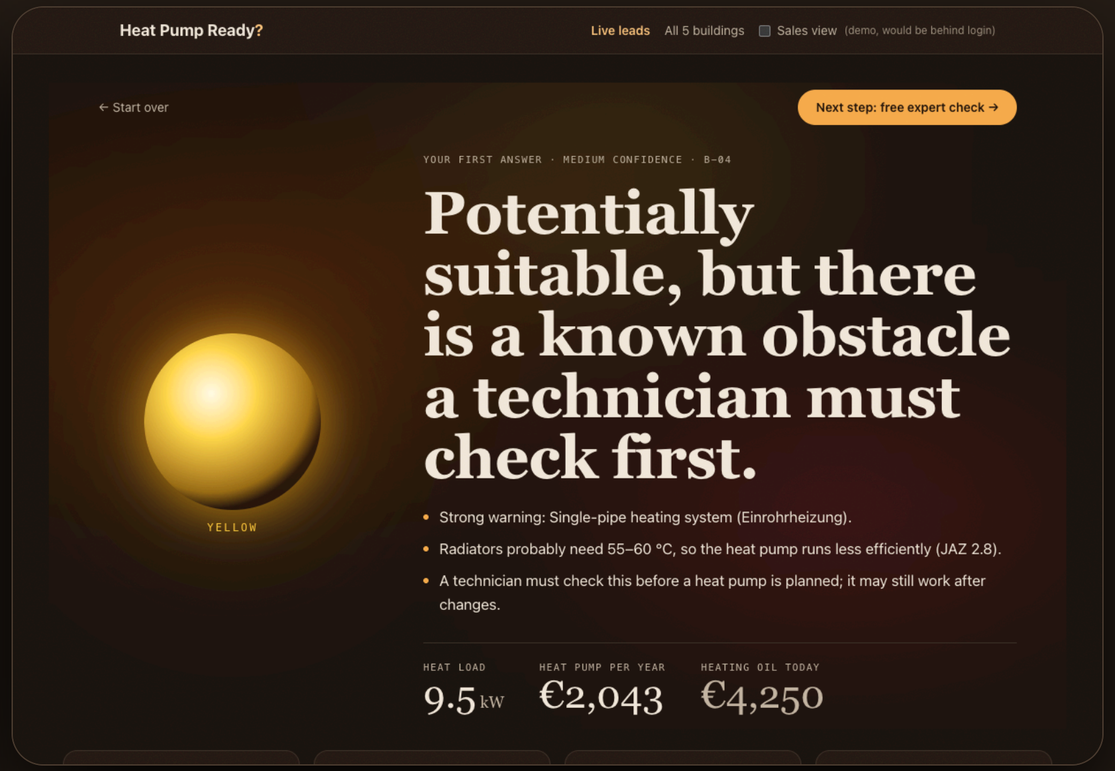
Not now

Info and subsidy guide; or outside the service area.

Expected value = chance of a sale × margin − visit cost, from inputs nuuEnergy sets live. Ranked by visible rules on building facts and the homeowner's own answers, never on age, income or personal traits.

A live thermal pre-scan, then *a verdict that glows.*



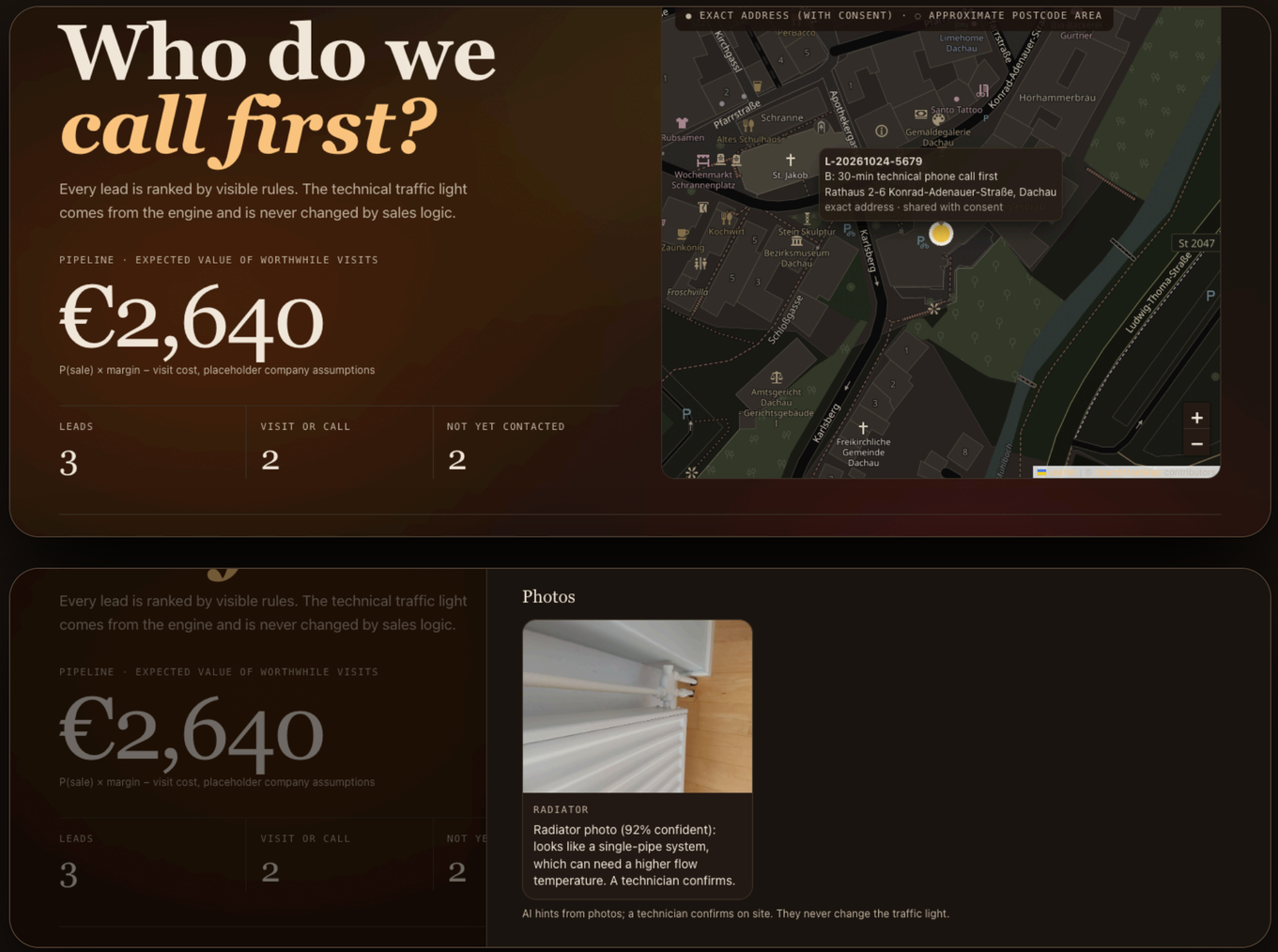


14 · THE SALES DESK

Who do we *call first?*

Live leads from the voice agent and the app, ranked, on a map. Click a lead and the map flies to the house; the drawer is a printable briefing for the heating master.

Exact building only with the customer’s consent; otherwise the postcode area.



Trust is the product.

Never asked, never scored

Age, income, health, nationality, bank details. The income bonus is a calculator on the homeowner’s device only.

Consent before anything moves

Details confirmed by the homeowner, then shared with consent; the right to object to profiling is stated.

Locked-down sales data

Leads, photos and addresses only behind a sales token; no keys in the app.

Scam-safe for older callers

Every call opens with: nuuEnergy will never ask for bank details or passwords.

From a challenge brief to *a working product.*

Morning
Engine on nuuEnergy's guardrails, web app, eval table

Midday
Sales tiers, website rules, subsidies, path to green

Afternoon
Agent backend, MCP tools, Astra orchestrator, voice

Evening
Photo AI, live leads, map, phone app, this deck

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155 automated tests passing

27 tools for the assistant

8 languages by voice

3 front ends, one engine

Ready to plug into *nuuEnergy's funnel.*

Learn from real outcomes

Log what each visit found, compare with the prediction, tune the rules. Still rules, never a black box.

Faster voice turns

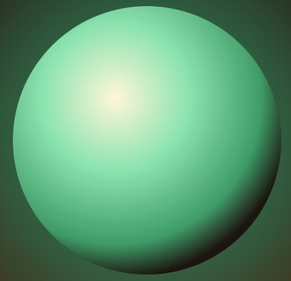
Today about 5–8 seconds per turn; streaming and caching bring it down.

Bigger photo test sets

nuuEnergy's own labelled photos turn our early signs into measured accuracy.

Into their CRM

Reports already go out as JSON, e-mail or webhook; the promises are settings, not code.



Fewer wasted visits. *More homes that say yes.*

Heat Pump Ready? · honest answers for homeowners, a smarter visit decision for nuuEnergy.

TEAM NOIDEA · WATT THE HACK MUNICH 2026 · THANK YOU